Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Investigation of Critical Heat Flux in Reduced Gravity using Photomicrographic Techniques		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	11/02/2002	**End Date:**	11/03/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2336		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effects of orientation on flow boiling critical heat flux (CHF) were investigated using high-speed video and microphotographic techniques. Interfacial features were measured just prior to CHF and statistically analyzed. A dominant wavy vapor layer regime was observed for all relatively high velocities and most orientations, while several other regimes were encountered at low velocities, in downflow and/or downward-facing heated wall orientations. The Interfacial Lift-off model was modified and used to predict the orientation effects on CHF for the dominant wavy vapor layer regime. The photographic study revealed a fairly continuous wavy vapor layer travelling along the heated wall while permitting liquid contact only in wetting fronts, located in the troughs of the interfacial waves. The waves, which were generated at an upstream location, had a tendency to preserve a curvature ratio as they propagated along the heated wall. CHF commenced when wetting fronts near the outlet were lifted off the wall. This occurred when the momentum of vapor normal to the wall exceeded the pressure force associated with interfacial curvature. The Interfacial		

Task Description:	Lift-off model is shown to be very effective at capturing the overall dependence of CHF on orientation. This study concerns the theoretical modeling of critical heat flux (CHF) in reduced gravity flow boiling. Throughout the study, the near-wall conditions at heat fluxes approaching and exceeding CHF will be examined with the aid of photomicrography. Key objectives of the proposed work are to (a) develop a test apparatus for exploring flow boiling and CHF in reduced gravity, (b) develop photographic methods for investigating vapor layer interfacial instabilities, (c) construct a phenomenological model for flow boiling CHF in reduced gravity, and (d) provide plans for testing of the same hardware in a future Space Shuttle follow-up study. The information provided from this project will be used to construct a phenomenological CHF model applicable to reduced gravity conditions which can be used as a design tool to guard against burnout of electronic and power devices in space thermal management systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 01/06/2026)